Question 01:

a) [0.25 each]

A) Agent
B) Sensors
C) Actuators
D) Percept's
E) Actions
F) Environment

b) [0.5 for the table and 0.5 for the agent selection]

Performance	**Environment**	**Actuators**	**Sensors**
Learning gain per session	Student's current answer	Problem selection engine	Student's text
Student engagement	Past performance history	Hint / explanation generator	Response time tracker
Problem completion rate	Curriculum knowledge graph	Difficulty adjuster	Error pattern analyser
Reduction in repeated errors	Session context (time of day, length)	Feedback display	Engagement signals (idle time)

Utility-based agent: A utility-based agent is preferable to a simple goal-based agent because there are competing desiderata: maximising learning gain sometimes conflicts with maintaining engagement.

c) **[0.4 each]**

A) Partially observable - Cameras and radar cannot see occluded vehicles; sensor range is finite; pedestrian intentions are hidden.
B) Stochastic - Unpredictable weather, other drivers' behaviour, and pedestrians mean that the same action does not always produce the same next state.
C) Sequential - Braking now affects the car's position and speed for all future decisions in the trip; each action has long-term consequences.
D) Dynamic - Other vehicles, pedestrians, and traffic signals all change while the agent is deliberating — the world does not wait.
E) Continuous - Speed, steering angle, and position are real-valued; precepts (camera images, radar distances) are continuous signals.

[0.5]

Multi-agent: Multiple autonomous vehicles from different companies operate simultaneously. This matters for algorithm choice because multi-agent settings may require game-theoretic reasoning or coordination protocols rather than single-agent planning.

Question 02: Part (a)

1. None of the variable domains change: 1 = {R, B} 2 = {R, B} 3 = {R, B} 4 = {R, B} 5 = {R, B}

2. 1 = {B} 2 = {R} 3 = {B} 4 = {R} 5 = {B}

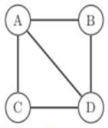
3. Forward checking is defined as a single iteration of constraint propagation only on those edges that terminate at the variable whose value was just set, and that do not originate from variables which have already been set. Therefore, after we set 1 = R, forward checking affects the domains of variables 2 and 4 since they are adjacent to variable 1 (and have not yet been assigned). 1 = {R} 2 = {B} 3 = {R, B} 4 = {B} 5 = {R, B}

4. Use of the most-constrained-variable strategy entails assigning the variable first whose domain is smallest. This ordering is not only performed at the start of the search. Rather, it is updated after each variable is assigned and forward checking modifies the domains of unassigned variables. For this problem, variable 5 has the smallest domain initially. After assigning variable 5, the domains of variable 2 and 4 become smaller than those of variables 3 and 5. Since variable 2 has the lowest index, it is assigned next. And so on, as shown below:

Assignment:	None	$5 = B$	$2 = R$	$1 = B$	$3 = B$	$4 = R$
Domain of 1:	$\{R, B\}$	$\{R, B\}$	$\{B\}$	**B**	**B**	**B**
Domain of 2:	$\{R, B\}$	$\{R\}$	**R**	**R**	**R**	**R**
Domain of 3:	$\{R, B\}$	$\{R, B\}$	$\{B\}$	$\{B\}$	**B**	**B**
Domain of 4:	$\{R, B\}$	$\{R\}$	$\{R\}$	$\{R\}$	$\{R\}$	**R**
Domain of 5:	$\{B\}$	**B**	**B**	**B**	**B**	**B**

Question 02: Part (b)

a)



b) Initial Domains

Before assignment:

A, B, C, D∈ {1,2,3}

After assigning:

C=2

Remaining domains:

A , B, D ∈ {1,2,3}

Apply Constraints Involving C

Constraint 1:

"If A and C live on the same floor, they must both be on floor 2."

Since $C = 2$:

$A = 2$is allowed.

No pruning yet.

Constraint 2:

"If A and C live on different floors, one of them must be on floor 3."

Since $C = 2$, if $A \neq 2$, then one of (A, C)must be 3.

But $C = 2$, so Amust be 3.

Therefore:

If $A \neq 2$, then $A = 3$

So $A = 1$becomes impossible.

Prune A:

$$A \in \{2,3\}$$

Constraint 3:

"D must live on a higher floor than C."

Since C = 2:

$$D > 2$$

Thus:

$$D = 3$$

Prune D:

$$D \in \{3\}$$

Constraint 4:

"D must not live on the same floor as anyone else."

Since $D = 3$:

No one else can be on floor 3.

So remove 3 from A and B.

Update A:

Current domain:

$$A \in \{2,3\}$$

Remove 3:

$$A \in \{2\}$$

Update B:

$$B \in \{1,2\}$$

Constraint 5:

"A and B must not live together."

Since A = 2:

$$B \neq 2$$

Thus:

$$B = 1$$

Person	Floor
A	2
B	1
C	2
D	3

Question 03: Part (a)

i) [0.5 each row]

F	C	
t	t	$0.1 \times 0.8 = 0.08$
t	f	$0.1 \times 0.2 = 0.02$
f	t	$0.9 \times 0.3 = 0.27$
f	f	$0.9 \times 0.7 = 0.63$

ii) [1 each]

$$P(C) = 0.08 + 0.27 = 0.35$$
$$P(F \mid C) = P(F, C) \,/\, P(C) = 0.08/0.35 \sim 0.23$$
$$P(F \mid \neg C) = P(F, \neg C) \,/\, P(\neg C) = 0.02/0.65 \sim 0.03$$

Question 03: Part (b)

Let R = {retirement plan}; H = {health insurance}. Given

$$P(R) = 0.56; \quad P(H) = 0.68; \quad P(RH) = 0.49.$$

(a) By Demorgan's rule

$$\begin{aligned} P(R^c \cap H^c) &= P(R \cup H)^c \\ &= 1 - P(R \cup H). \end{aligned}$$

Now,

$$\begin{aligned} P(R \cup H) &= P(R) + P(H) - P(R \cap H) \\ &= 0.56 + 0.68 - 0.49 = 0.75. \end{aligned}$$

Therefore,

$$P(R^c \cap H^c) = 1 - (0.75) = 0.25.$$

(b) Also,

$$\begin{aligned} P(H|R) &= \frac{P(H \cap R)}{P(R)} \\ &= \frac{0.49}{0.56} = \frac{7}{8}. \end{aligned}$$

(c) Since $P(H|R) = \frac{7}{8} \neq P(H) = 0.68$, the events H and R are not independent.

(d) Are H and R mutually exclusive?

Since $P(HR) = 0.49 > 0$, the events H and R are not mutually exclusive.

Question 04: Part (a)

1)We need a train–test split to properly evaluate how well a machine learning model will perform on new, unseen data. If we use all the data for training, the model may simply memorize patterns (overfitting) and appear very accurate, but fail in real-world predictions. So we split data into: • Training set: used to learn patterns and build the model • Test set: used to check how well the model generalizes to unseen data

2) We sometimes need an additional validation dataset because using only a train-test split is not enough for proper model development

3) a)

	Predicted On Time	Predicted Delayed
Actual On Time	6 (TP)	4 (FN)
Actual Delayed	1 (FP)	5 (TN)

b) **Accuracy**

$$\frac{TP+TN}{Total} = \frac{6+5}{16} = \frac{11}{16} = 0.6875$$

Accuracy = 68.75%
Precision (On Time)

$$\frac{TP}{TP+FP} = \frac{6}{6+1} = \frac{6}{7} \approx 0.857$$

Precision = 85.7%
Recall (On Time)

$$\frac{TP}{TP+FN} = \frac{6}{6+4} = \frac{6}{10} = 0.6$$

Recall = 60%
F1 Score

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$
$$= 2 \cdot \frac{0.857 \cdot 0.6}{0.857 + 0.6} \approx 0.706$$

F1 Score = 70.6%
d) Recall is more important
e) If the model predicts a positive class it is likely to be **correct**. However, for classifying positive instances correctly, the model performs similarly to **precision** and **recall**. We can conclude the first insight from the **precision** and the second from the **recall**.

Question 04: Part (b)

Solution

a) $z = \beta_0 + \beta_1 \times X$, $z = 5.40 + (-0.09 \times 45)$, $z = 5.40 + (-4.05)$ $z = 5.40 - 4.05$, $z = \mathbf{+1.35}$ and $z = \beta_0 + \beta_1 \times X$, $z = 5.40 + (-0.09 \times 70)$, $z = 5.40 + (-6.30)$, $z = 5.40 - 6.30$, $z = \mathbf{-0.90}$

b) $-z = -(+1.35) = \mathbf{-1.35}$, $e^{-1.35} = 1 / e^{1.35}$, $e^{1.35} \approx e^{1} \times e^{0.35} \approx 2.71828 \times 1.4191 \approx 3.8574$, $e^{-1.35} = \mathbf{1 / 3.8574 \approx 0.2592}$, $P = 1 / (1 + e^{-z}) = 1 / (1 + 0.2592) = 1 / 1.2592 = \mathbf{0.7942}$ and $-z = -(-0.90) = +0.90$, $e^{+0.90} \approx 2.4596$, $P = 1 / (1 + e^{-z}) = 1 / (1 + e^{+0.90}) = 1 / (1 + 2.4596) = 1 / 3.4596 = \mathbf{0.2890}$

c) P = 0.7942 Is 0.7942 ≥ 0.50? = Yes, predicted class = **1 (will default)** Bank decision = **reject loan application** and P = 0.2890 Is 0.2890 ≥ 0.50? = No, predicted class = **0 (will not default)** Bank decision = **approve loan application**

Question 05:

Solution:
a. Entropy of dataset
Yes = 3, No = 4
$\text{Entropy}(S) = H(S) = -\frac{3}{7}\log_2\frac{3}{7} - \frac{4}{7}\log_2\frac{4}{7} = 0.985$

b. Information gain for each attribute

- **Fiction**
  no: 1Y,1N → H=1
  yes: 2Y,3N → H=0.971
  Weighted = (2/7)*1 + (5/7)*0.971 = 0.979
  Gain = 0.985 - 0.979 = **0.006**

- **Age**
  new: 2Y,1N → H=0.918
  ancient: 0Y,2N → H=0
  old: 1Y,1N → H=1

Weighted = (3/7)*0.918 + (2/7)*0 + (2/7)*1 = 0.679
Gain = 0.985 - 0.679 = **0.306**

- **Award**
  no: 2Y,3N → H=0.971
  yes: 1Y,1N → H=1
  Weighted = (5/7)*0.971 + (2/7)*1 = 0.979
  Gain = 0.985 - 0.979 = **0.006**

- **Rating**
  hi: 3Y,0N → H=0
  med: 0Y,2N → H=0
  low: 0Y,2N → H=0
  Weighted = 0
  Gain = 0.985 - 0 = **0.985**

- **Length**
  short: 2Y,2N → H=1
  long: 0Y,1N → H=0
  medium: 1Y,1N → H=1
  Weighted = (4/7)*1 + (1/7)*0 + (2/7)*1 = 0.857
  Gain = 0.985 - 0.857 = **0.128**

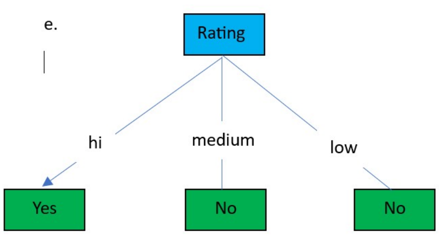
c. Root node selection

Attribute	Gain
Rating	**0.985**
Age	0.306
Length	0.128
Fiction	0.006
Award	0.006

Best attribute: Rating

d. All three values of Rating map to only one value of Read, meaning Rating alone can determine the value of Read. All other attributes have at least two values that have mixed Read values.

e.



Question 06: Part (a)

a) Euclidean Distance Calculations

ID 1: (2,10)
d = √[(4−2)^2 + (10−10)^2] = √(4+0) = 2
Distance = 2.00

ID 7: (3,8)
d = √[(4−3)^2 + (10−8)^2] = √(1+4) = √5 = 2.236
Distance ≈ 2.24

ID 8: (2,4)
d = √[(4−2)^2 + (10−4)^2] = √(4+36) = √40 = 6.325
Distance ≈ 6.32

b) KNN Classification (K=3)

ID	(X1,X2)	Distance	Class
1	(2,10)	2.00	Yes
2	(1,5)	5.83	No
3	(5,20)	10.05	Yes
4	(8,15)	6.40	Yes
5	(1,2)	8.54	No
6	(10,30)	20.88	No
7	(3,8)	2.24	Yes
8	(2,4)	6.32	No
9	(4,25)	15.00	Yes
10	(6,12)	2.83	Yes

Three nearest neighbors: ID1, ID7, ID10
Classes: Yes, Yes, Yes
Predicted Credit Approval (Y): YES

Question 06: Part (b)
a)
$\Sigma X = 10 + 20 + 30 + 40 + 50 + 60 = 210$
$\Sigma Y = 25 + 45 + 60 + 80 + 95 + 115 = 420$
$\bar{x} = \Sigma X / n = 210 / 6 = 35$
$\bar{y} = \Sigma Y / n = 420 / 6 = 70$

b) Calculate Required Columns

X	Y	$(x_i-\bar{x})$	$(y_i-\bar{y})$	$(x_i-\bar{x})(y_i-\bar{y})$	$(x_i-\bar{x})^2$
10	25	-25	-45	1125	625
20	45	-15	-25	375	225
30	60	-5	-10	50	25
40	80	5	10	50	25
50	95	15	25	375	225
60	115	25	45	1125	625

$\Sigma(x_i-\bar{x})(y_i-\bar{y}) = 3100$
$\Sigma(x_i-\bar{x})^2 = 1750$

c) Calculate Slope (m) and Intercept (b)

$m = 3100 / 1750 = 1.771$
$b = \bar{y} - m\,\bar{x} = 70 - (1.771 \times 35) = 8.015$
Regression Equation: $y = 1.771x + 8.015$

d) Predict July Sales (X = 75)

$y = 1.771(75) + 8.015$
$= 132.825 + 8.015$
$= 140.84$
Predicted Sales Revenue ≈ 141 PKR thousand.

Question 06: Part (c)

1. Distance table

Point	(X_1,X_2)	$d(P, C_1)$	$d(P, C_2)$	Assigned to
P1	(1,1)	$\sqrt{[(1-1)^2+(1-1)^2]} = 0.00$	$\sqrt{[(1-8)^2+(1-7)^2]} = \sqrt{85} \approx 9.22$	C_1
P2	(1,3)	$\sqrt{[(1-1)^2+(3-1)^2]} = \sqrt{4} = 2.00$	$\sqrt{[(1-8)^2+(3-7)^2]} = \sqrt{65} \approx 8.06$	C_1
P3	(2,2)	$\sqrt{[(2-1)^2+(2-1)^2]} = \sqrt{2} \approx 1.41$	$\sqrt{[(2-8)^2+(2-7)^2]} = \sqrt{61} \approx 7.81$	C_1
P4	(8,7)	$\sqrt{[(8-1)^2+(7-1)^2]} = \sqrt{85} \approx 9.22$	$\sqrt{[(8-8)^2+(7-7)^2]} = 0.00$	C_2
P5	(9,8)	$\sqrt{[(9-1)^2+(8-1)^2]} = \sqrt{113} \approx 10.63$	$\sqrt{[(9-8)^2+(8-7)^2]} = \sqrt{2} \approx 1.41$	C_2
P6	(8,9)	$\sqrt{[(8-1)^2+(9-1)^2]} = \sqrt{113} \approx 10.63$	$\sqrt{[(8-8)^2+(9-7)^2]} = \sqrt{4} = 2.00$	C_2

Cluster 1 (C_1): P1, P2, P3   Cluster 2 (C_2): P4, P5, P6
Compute new centroids after Iteration 1:New $C_1 = ((1+1+2)/3 , (1+3+2)/3) = (4/3, 6/3) = (1.33, 2.00)$
New $C_2 = ((8+9+8)/3 , (7+8+9)/3) = (25/3, 24/3) = (8.33, 8.00)$

2. Iteration 2 — Reassign with C_1=(1.33, 2.00), C_2=(8.33, 8.00)

Point	(X_1,X_2)	$d(P, C_1)$	$d(P, C_2)$	Assigned to
P1	(1,1)	$\sqrt{[(1-1.33)^2+(1-2)^2]} \approx 1.05$	$\sqrt{[(1-8.33)^2+(1-8)^2]} \approx 10.26$	C_1
P2	(1,3)	$\sqrt{[(1-1.33)^2+(3-2)^2]} \approx 1.05$	$\sqrt{[(1-8.33)^2+(3-8)^2]} \approx 8.92$	C_1
P3	(2,2)	$\sqrt{[(2-1.33)^2+(2-2)^2]} \approx 0.67$	$\sqrt{[(2-8.33)^2+(2-8)^2]} \approx 8.82$	C_1
P4	(8,7)	$\sqrt{[(8-1.33)^2+(7-2)^2]} \approx 8.42$	$\sqrt{[(8-8.33)^2+(7-8)^2]} \approx 1.05$	C_2
P5	(9,8)	$\sqrt{[(9-1.33)^2+(8-2)^2]} \approx 9.78$	$\sqrt{[(9-8.33)^2+(8-8)^2]} \approx 0.67$	C_2
P6	(8,9)	$\sqrt{[(8-1.33)^2+(9-2)^2]} \approx 9.55$	$\sqrt{[(8-8.33)^2+(9-8)^2]} \approx 1.05$	C_2

Compute centroids after Iteration 2: C_1 = (1.33, 2.00) — same clusters as before: C_2 = (8.33, 8.00) — same clusters as before

Final Cluster 1: P1(1,1), P2(1,3), P3(2,2) — centroid (1.33, 2.00)
Final Cluster 2: P4(8,7), P5(9,8), P6(8,9) — centroid (8.33, 8.00)

Question 07 Part (a)

Node N3
$z3=(1)(-0.8)+(-0.5)(-0.6)+(-0.5)$
$=-0.8+0.3-0.5=-1.0$
$N3=\sigma(-1)=1/(1+e^1)=0.2689$
Node N4
Weights: w14=0.5, w24=0.4, b4=-0.3
$z4=(1)(0.5)+(-0.5)(0.4)-0.3$
$=0.5-0.2-0.3=0$
$N4=\sigma(0)=0.5$
Node N5
Weights: w35=-0.9, w45=0.3, b5=-0.4
$z5=(0.2689)(-0.9)+(0.5)(0.3)-0.4$
$=-0.2420+0.15-0.4=-0.4920$
$N5=\sigma(-0.4920)=0.3794$
Node N6
Weights: w36=0.7, w46=0.2, b6=-0.2
$z6=(0.2689)(0.7)+(0.5)(0.2)-0.2$
$=0.1883+0.1-0.2=0.0883$
$N6=\sigma(0.0883)=0.5221$
Node N7
Weights: w57=-0.8, w67=-0.6, b7=-1.5
$z7=(0.3794)(-0.8)+(0.5221)(-0.6)-1.5$
$=-0.3035-0.3133-1.5=-2.1168$
$N7=\sigma(-2.1168)=0.1075$

Question 07 Part (b) (Partial Solution) (Please complete the left over at your own)

Output Layer Error

$\delta7=(T-O7)O7(1-O7)$
$=(0.5-0.1075)(0.1075)(1-0.1075)=0.03765$

2. Hidden Layer Deltas

$\delta5=(0.3794)(0.6206)(-0.8)(0.03765)=-0.00709$
$\delta6=(0.5221)(0.4779)(-0.6)(0.03765)=-0.00564$

3. Weight Updates

Formula: w_new = w_old + ηδ(input)
$\Delta w57=0.5(0.03765)(0.3794)=0.00714$
$w57,new = -0.8 + 0.00714 = -0.7929$
$\Delta w67=0.5(0.03765)(0.5221)=0.00983$
$w67,new = -0.6 + 0.00983 = -0.5902$
$\Delta b7=0.5(0.03765)=0.01883$
$b7,new = -1.5 + 0.01883 = -1.4812$
$\Delta w35=0.5(-0.00709)(0.2689)=-0.00095$
$w35,new = -0.90095$
$\Delta w45=0.5(-0.00709)(0.5)=-0.00177$
$w45,new = 0.2982$

Final Updated Parameters

Parameter	Updated Value
w57	−0.7929
w67	−0.5902
b7	−1.4812
w35	−0.90095
w45	0.2982

Question 08

1. $Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [R + \gamma \cdot \max Q(s', a')]$

$Q(s, a)$= Current Q-value for state s, action a
α (alpha)+ Learning rate , how fast the agent updates its knowledge (0 to 1)
R= Immediate reward received after taking action a in state s
γ (gamma)= Discount factor, how much future rewards are valued (0 to 1)
$\max Q(s', a')$= Best Q-value available from the next state s'

2. $Q(1,2, Up) \leftarrow 0 + 0.5 \times [100 + 0.9 \times 0]$

3. Solve inside the bracket: $= 100 + (0.9 \times 0) =100 + 0 = 100$

Now: Multiply by α: $0.5 \times 100 = 50$
Add to current Q:
$Q(1,2, Up) \leftarrow 0 + 50 = 50$

4. $R = 100$, $Q(0,1,Up) = 0$, $\alpha = 0.5$, $\gamma = 0.9$, $\max Q(s',a') = 50$ (from part c)

$Q(0,1, Up) \leftarrow 0 + 0.5 \times [100 + 0.9 \times 50]$
Solve inside the bracket: $= 100 + (0.9 \times 50) = 100 + 45 = 145$
Now Multiply by α: $0.5 \times 145 = 72.5$
Add to current Q: $Q(0,1, Up) \leftarrow 0 + 72.5 = 72.5$